OPERATIONS MANAGEMENT

CAPACITY PLANNING

ARJUN SINGAR

Capacity Management

- **Capacity** is the maximum rate of output of a process or a system.

- Managers are responsible for ensuring that the firm has the capacity to meet current and future demand. Otherwise, the organization will miss out on opportunities for growth and profits.

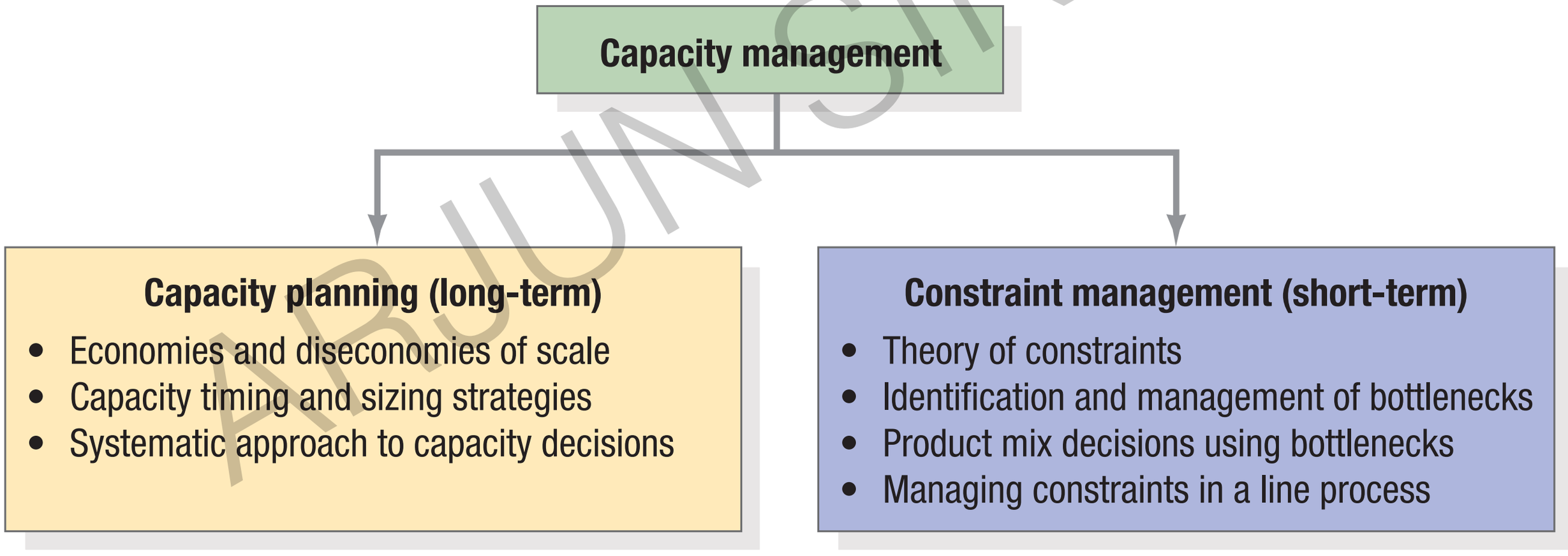
- Increasing or decreasing capacity by itself is not as important as ensuring that the entire supply chain, from order entry to delivery, is designed for effectiveness.

- Capacity decisions must be made in light of several long-term issues such as the firm's economies and diseconomies of scale, capacity cushions, timing and sizing strategies, and trade-offs between customer service and capacity utilization.

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- This chapter focuses on how managers can best revise capacity levels and best determine when to add or reduce capacity for the long term.

- Both long-term as well as short-term issues associated with planning capacity and managing constraints are important, and must be understood in conjunction with one another.



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Planning Long-Term Capacity

- Long-term capacity plans deal with investments in new facilities and equipment at the organizational level, and require top management participation and approval because they are not easily reversed.

- When choosing a capacity strategy, managers must consider questions such as the following: How much of a cushion is needed to handle variable, or uncertain, demand?

- Should we expand capacity ahead of demand, or wait until demand is more certain?

- Even before these questions can be answered, a manager needs to be able to measure a process's capacity.

- So a systematic approach is needed to answer these and similar questions and to develop a capacity strategy appropriate for each situation.

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Measures of Capacity and Utilization

- **Output Measures of Capacity** are best utilized when applied to individual processes within the firm, or when the firm provides a relatively small number of standardized services and products. High-volume processes, such as those in a car manufacturing plant, are a good example. In this case, capacity would be measured in terms of the number of cars produced per day.

- **Input Measures of Capacity** are generally used for low-volume, flexible processes, such as those associated with a custom furniture maker. In this case, the furniture maker might measure capacity in terms of inputs such as number of workstations or number of workers required to fulfill the demand.

- **Utilization** is the degree to which a resource such as equipment, space, or the workforce is currently being used, and is measured as the **ratio of average output rate to maximum capacity** (expressed as a percent). The average output rate and the capacity must be measured in the same terms—that is, time, customers, units, or dollars. The utilization rate indicates the need for adding extra capacity or eliminating unneeded capacity.

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Economies of Scale

- A concept known as economies of scale states that the average unit cost of a service or good can be reduced by increasing its output rate. Four principal reasons explain why economies of scale can drive costs down when output increases: (1) Fixed costs are spread over more units; (2) construction costs are reduced; (3) costs of purchased materials are cut; and (4) process advantages are found.

- **Spreading Fixed Costs:** These fixed costs include equipment costs, power costs, and managers' salaries. When the average output rate increases, the average unit cost drops because fixed costs are spread over more units.

- **Reducing Construction Costs:** Certain activities and expenses are required to build small and large facilities alike - building permits, architects' fees, and rental of building equipment. Doubling the size of the existing facility usually does not double construction costs.

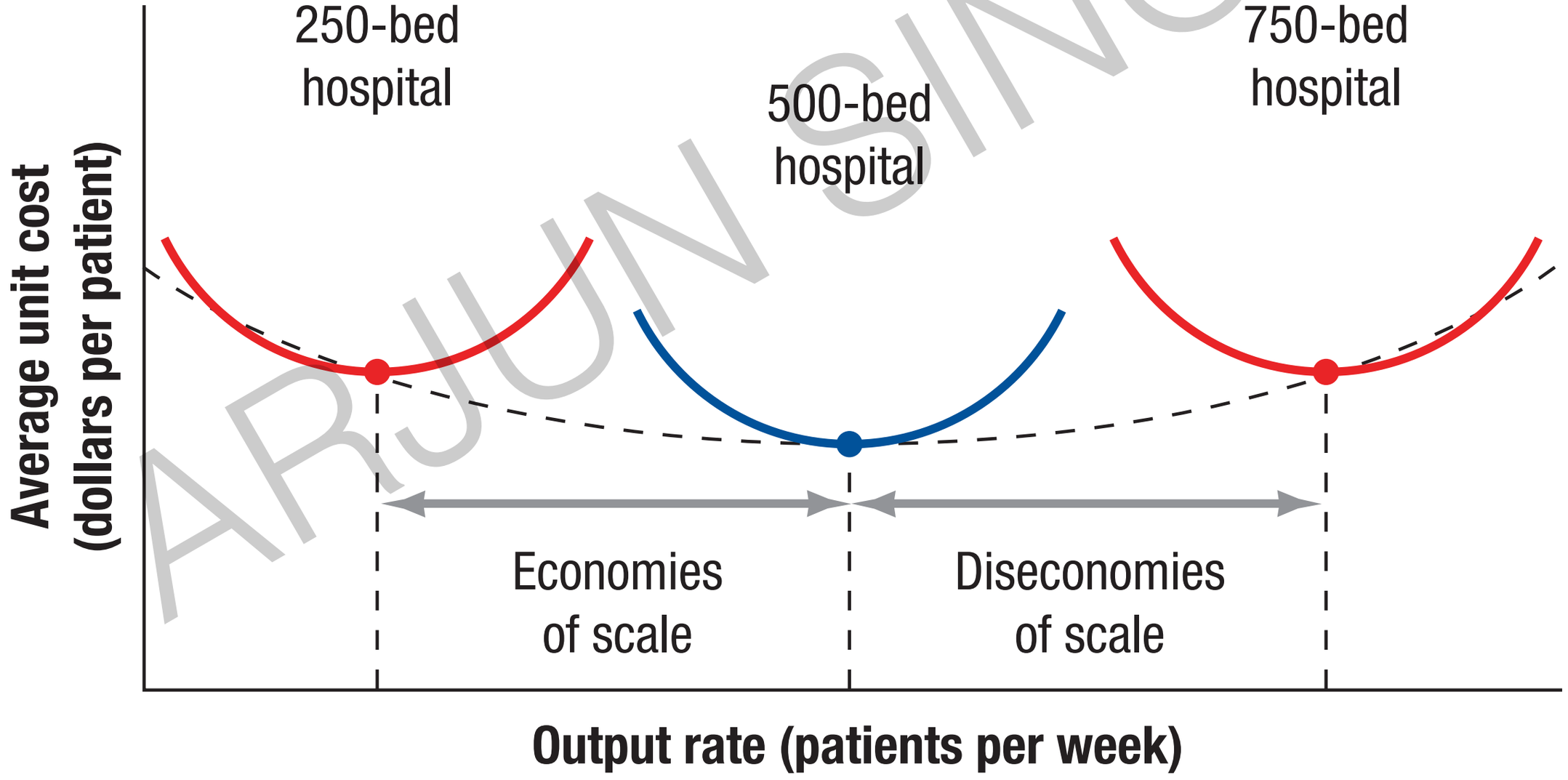
- **Cutting Costs of Purchased Materials:** Higher volumes can reduce the costs of purchased materials and services. They give the purchaser a better bargaining position and the opportunity to take advantage of quantity discounts.

- **Finding Process Advantages:** High-volume production provides many opportunities for cost reduction. At a higher output rate, the process shifts toward a line process, with resources dedicated to individual products. The benefits from dedicating resources to individual services or products may include speeding up the learning effect, lowering inventory, improving process and job designs, and reducing the number of changeovers.

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Diseconomies of Scale

- Bigger is not always better, however; at some point, a facility can become so large that **diseconomies of scale** set in; that is, the average cost per unit increases as the facility's size increases. The reason is that excessive size can bring complexity, loss of focus, and inefficiencies that raise the average unit cost of a service or product.



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Capacity Timing and Sizing Strategies

- Operations managers must examine three dimensions of capacity strategy before making capacity decisions: (1) sizing capacity cushions, (2) timing and sizing expansion, and (3) linking process capacity and other operating decisions.

Sizing Capacity Cushions

- Average utilization rates for any resource should not get too close to 100 percent over the long term, though it may occur for some processes from time to time in the short run. If the demand keeps increasing over time, then long-term capacity must be increased as well to provide some buffer against uncertainties. When average utilization rates approach 100 percent, it is usually a signal to increase capacity or decrease order acceptance to avoid declining productivity. The **capacity cushion** is the amount of reserve capacity a process uses to handle sudden increases in demand or temporary losses of production capacity; it measures the amount by which the average utilization (in terms of total capacity) falls below 100 percent.

Capacity cushion = 100 (%) - Average Utilization rate (%)

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- For capital-intensive firms, minimizing the capacity cushion is vital. Studies indicate that businesses with high capital intensity achieve a low return on investment when the capacity cushion is high. This strong correlation does not exist for labor-intensive firms; however, their return on investment is about the same because the lower investment in equipment makes high utilization less critical.

- Advantages of small cushion are lesser unused capacity and detect inefficiencies such as employee absenteeism and unreliable suppliers.

- The appropriate size of the cushion varies by industry. In the capital-intensive paper industry, where machines can cost hundreds of millions of dollars each, cushions well under 10 percent are preferred. The less capital-intensive hotel industry breaks even with a 60 to 70 percent utilization (40 to 30 percent cushion), and begins to suffer customer-service problems when the cushion drops to 20 percent.

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Timing and Sizing Expansion

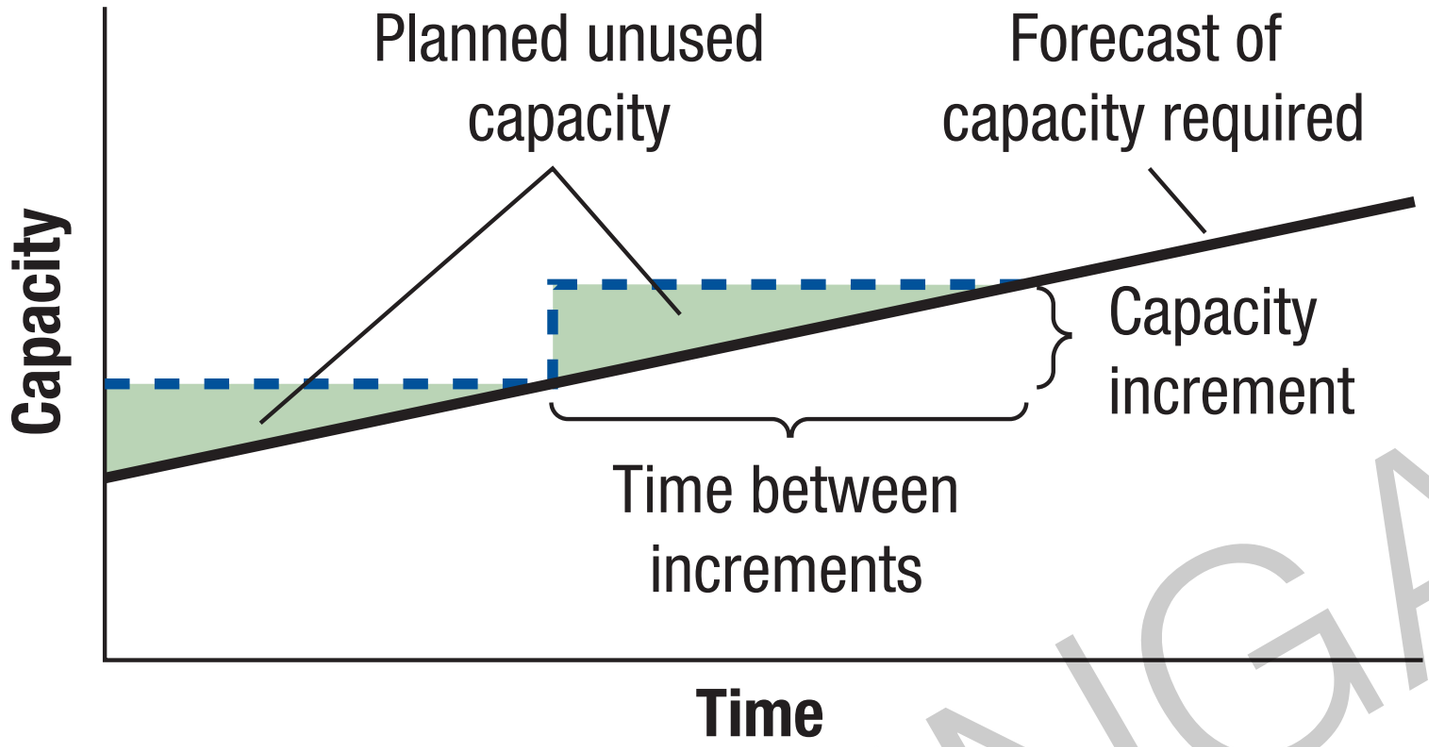
- The second issue of capacity strategy concerns when to adjust capacity levels and by how much. At times, capacity expansion can be done in response to changing market trends.

There are two extreme strategies for expanding capacity:

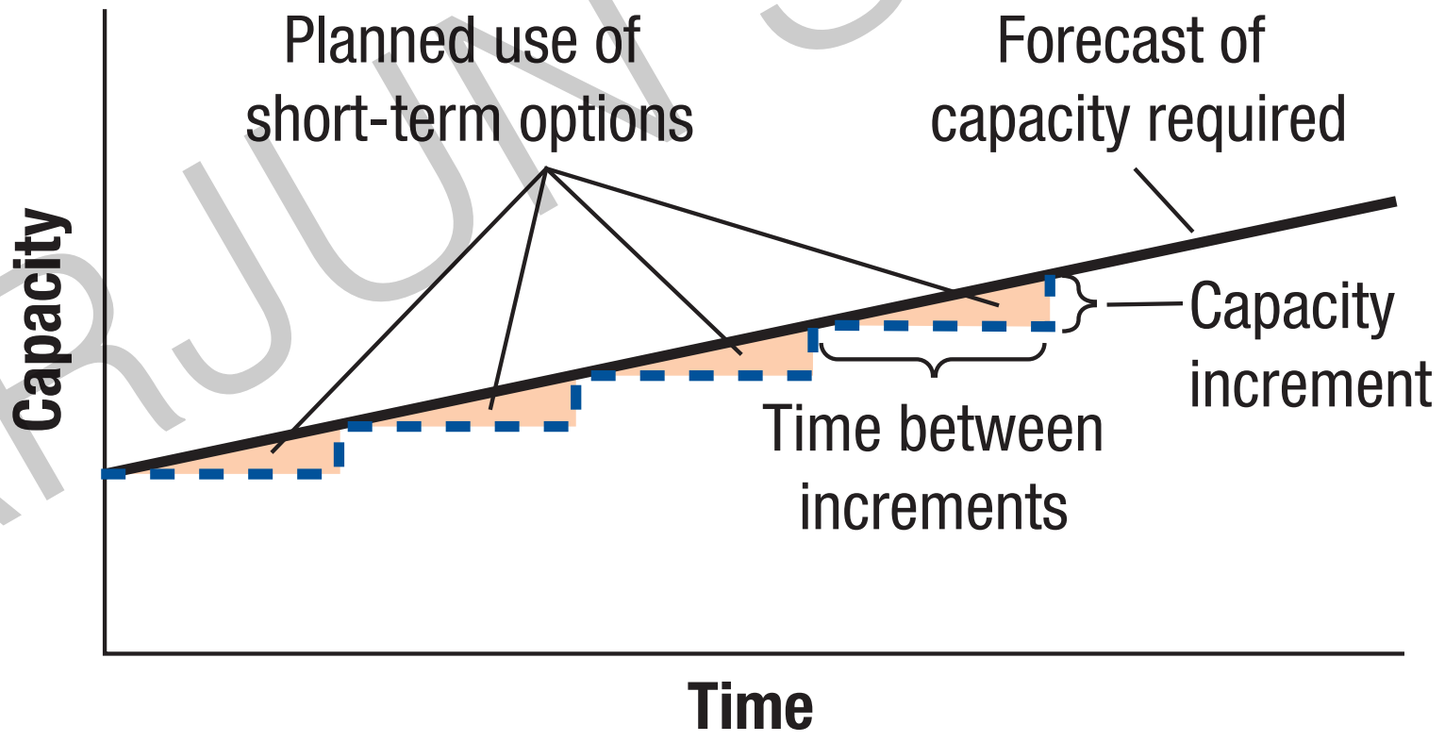
- The ***expansionist strategy*** - which involves large, infrequent jumps in capacity. The timing and sizing of expansion are related; that is, if demand is increasing and the time between increments increases, the size of the increments must also increase. The expansionist strategy, which stays ahead of demand, minimizes the chance of sales lost to insufficient capacity. Several factors favor the expansionist strategy. Expansion can result in economies of scale and a faster rate of learning, thus helping a firm reduce its costs and compete on price.

- The ***wait-and-see strategy*** - which involves smaller, more frequent jumps. The wait-and-see strategy lags behind demand. To meet any shortfalls, it relies on short-term options, such as use of overtime, temporary workers, subcontractors, and the postponement of preventive maintenance on equipment. The conservative wait-and-see strategy is to expand in smaller increments, such as by renovating existing facilities rather than building new ones. Because the wait-and-see strategy follows demand, it reduces the risks of over-expansion based on overly optimistic inaccurate demand forecasts, or obsolete technology.

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(a) Expansionist strategy



(b) Wait-and-see strategy

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A Systematic Approach to Long-Term Capacity Decisions

- Long-term decisions for capacity would typically include whether to add a new plant or ware-house or to reduce the number of existing ones, how many workstations a given department should have, or how many workers are needed to staff a given process.

- Some of these decisions can take years to become operational. Hence, a systematic approach is needed to plan for long- term capacity decisions.

- Although each situation is somewhat different, a four-step procedure generally can help managers make sound capacity decisions:

1. Estimate future capacity requirements.

2. Identify gaps by comparing requirements with available capacity.

3. Develop alternative plans for reducing the gaps.

4. Evaluate each alternative, both qualitatively and quantitatively, and make a final choice.

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Estimate Capacity Requirements

- A process's **capacity requirement** is what its capacity should be for some future time period to meet the demand of the firm's customers (external or internal), given the firm's desired capacity cushion.

- Capacity requirements can be expressed in one of two ways: with an output measure or with an input measure. Either way, the foundation for the estimate is forecasts of demand, productivity, competition, and technological change. These forecasts normally need to be made for several time periods in a **planning horizon**, which is the set of consecutive time periods considered for planning purposes.

- **Using Output Measures** - The simplest way to express capacity requirements is as an output rate. Output measures are appropriate for high-volume processes with little product variety or process divergence. Here, demand forecasts for future years are used as a basis for extrapolating capacity requirements into the future. For example, if a process's current demand is 50 customers per day, then the demand in five years would be 100 customers per day. If the desired capacity cushion is 20 percent, management should plan for enough capacity to serve [100/(1 - 0.2)] = 125 customers in five years.

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- **Using Input Measures** - Output measures may be insufficient in the following situations: (i) Product variety and process divergence is high, (ii) The product or service mix is changing, (iii) Productivity rates are expected to change, (iv) Significant learning effects are expected.

- In such cases, it is more appropriate to calculate capacity requirements using an input measure, such as the number of employees, machines, computers, or trucks. Using an input measure for the capacity requirement brings together demand forecasts, process time estimates, and the desired capacity cushion. When just one service or product is processed at an operation and the time period is a particular year, the capacity requirement, *M*, is

$$\text{Capacity requirement} = \frac{\text{Processing hours required for year's demand}}{\text{Hours available from a single capacity unit (such as an employee or machine) per year, after deducting desired cushion}}$$

$$M = \frac{Dp}{N[1 - (C/100)]}$$

where

D = demand forecast for the year (number of customers served or units produced)

p = processing time (in hours per customer served or unit produced)

N = total number of hours per year during which the process operates

C = desired capacity cushion (expressed as a percent)

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- *M* is the number of input units required and should be calculated for each year in the time horizon. The processing time, *p*, depends on the process and methods selected to do the work. The denominator is the total number of hours, *N*, available for the year from one unit of capacity (an employee or machine), multiplied by a proportion that accounts for the desired capacity cushion, *C.* The proportion is simply 1.0 - *C*/100, where *C* is converted from a percent to a proportion by dividing by 100. For example, a 20 percent capacity cushion means that 1.0 - *C*/100 = 0.80.

- Setups may be involved if multiple products are being manufactured. **Setup time** is the time required to change a process or an operation from making one service or product to making another. The total setup time is found by dividing the number of units forecast per year, *D*, by the number of units made in each lot, *Q*, (number of units processed between setups), which gives the number of setups per year, and then multiplying by the time per setup, *s*. For example, if the annual demand is 1,200 units and the average lot size is 100, there are 1,200/100 = 12 setups per year. Accounting for both processing and setup times for multiple services (products), we get

$$\text{Capacity requirement} = \frac{\begin{array}{c}\text{Processing and setup hours required for}\\ \text{year's demand, summed over all services or products}\end{array}}{\begin{array}{c}\text{Hours available from a single capacity unit per year,}\\ \text{after deducting desired cushion}\end{array}}$$

$$M = \frac{[Dp + (D/Q)s]_{\text{product 1}} + [Dp + (D/Q)s]_{\text{product 2}} + \cdots + [Dp + (D/Q)s]_{\text{product } n}}{N\,[1-(C/100)]}$$

where

Q = number of units in each lot

s = setup time (in hours) per lot

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Identifying Gaps

- A **capacity gap** is any difference (positive or negative) between projected capacity requirements (*M*) and current capacity. Complications arise when multiple operations and several resource inputs are involved. Expanding the capacity of some operations may increase overall capacity.

Develop Alternatives

- The next step is to develop alternative plans to cope with projected gaps. One alternative, called the **base case**, is to do nothing and simply lose orders from any demand that exceeds current capacity or incur costs because capacity is too large. Other alternatives if expected demand exceeds current capacity are various timing and sizing options for adding new capacity.

Evaluate the Alternatives

- **Qualitative Concerns** - Qualitatively, the manager looks at how each alternative fits the overall capacity strategy and other aspects of the business not covered by the financial analysis. Of particular concern might be uncertainties about demand, competitive reaction, technological change, and cost estimates. Some of these factors cannot be quantified and must be assessed on the basis of judgment and experience.

- **Quantitative Concerns** - Quantitatively, the manager estimates the change in cash flows for each alternative over the forecast time horizon compared to the base case. **Cash flow** is the difference between the flows of funds into and out of an organization over a period of time, including revenues, costs, and changes in assets and liabilities.

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Application

- A copy center in an office building prepares bound reports for two clients. The center makes multiple copies (the lot size) of each report. The processing time to run, collate, and bind each copy depends on, among other factors, the number of pages. The center operates 250 days per year, with one 8-hour shift. Management believes that a capacity cushion of 15 percent (beyond the allowance built into time standards) is best. It currently has three copy machines. Based on the following table of information, determine how many machines are needed at the copy center.

Item	Client X	Client Y
Annual demand forecast (copies)	**2,000**	**6,000**
Standard processing time (hour/copy)	**0.5**	**0.7**
Average lot size (copies per report)	**20**	**30**
Standard setup time (hours)	**0.25**	**0.40**

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Solution

$$M = \frac{[Dp + (D/Q)s]_{\text{product 1}} + [Dp + (D/Q)s]_{\text{product 1}} + \ldots + [Dp + (D/Q)s]_{\text{product } n}}{N[1 - (C/100)]}$$

$$= \frac{[2{,}000(0.5) + (2{,}000/20)(0.25)]_{\text{client X}} + [6{,}000(0.7) + (6{,}000/30)(0.40)]_{\text{client Y}}}{[(250 \text{ day/year})(1 \text{ shift/day})(8 \text{ hours/shift})][1.0 - (15/100)]}$$

$$= \frac{5{,}305}{1{,}700} = 3.12$$

Rounding up to the next integer gives a requirement of four machines.

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Application

A Grandmother's Chicken Restaurant is experiencing a boom in business. The owner expects to serve 80,000 meals this year. Although the kitchen is operating at 100 percent capacity, the dining room can handle 105,000 diners per year. Forecasted demand for the next five years is 90,000 meals for next year, followed by a 10,000-meal increase in each of the succeeding years. One alternative is to expand both the kitchen and the dining room now, bringing their capacities up to 130,000 meals per year. The initial investment would be $200,000, made at the end of this year (year 0). The average meal is priced at $10, and the before-tax profit margin is 20 percent. The 20 percent figure was arrived at by determining that, for each $10 meal, $8 covers variable costs and the remaining $2 goes to pretax profit.

What are the pretax cash flows from this project for the next five years compared to those of the base case of doing nothing?

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Solution

Recall that the base case of doing nothing results in losing all potential sales beyond 80,000 meals. With the new capacity, the cash flow would equal the extra meals served by having a 130,000-meal capacity, multiplied by a profit of $2 per meal. In year 0, the only cash flow is –$200,000 for the initial investment. In year 1, the 90,000-meal demand will be completely satisfied by the expanded capacity, so the incremental cash flow is (90,000 – 80,000)($2) = $20,000. For subsequent years, the figures are as follows:

Year 2: Demand = 100,000; Cash flow = (100,000 – 80,000)$2 = $40,000

Year 3: Demand = 110,000; Cash flow = (110,000 – 80,000)$2 = $60,000

Year 4: Demand = 120,000; Cash flow = (120,000 – 80,000)$2 = $80,000

Year 5: Demand = 130,000; Cash flow = (130,000 – 80,000)$2 = $100,000

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Solution

If the new capacity were smaller than the expected demand in any year, we would subtract the base case capacity from the new capacity (rather than the demand). The owner should account for the time value of money, applying such techniques as the net present value or internal rate of return methods. For instance, the net present value (NPV) of this project at a discount rate of 10 percent is calculated here, and equals $13,051.76.

$$NPV = -200{,}000 + [(20{,}000/1.1)] + [40{,}000/(1.1)^2] + [60{,}000/(1.1)^3] + [80{,}000/(1.1)^4] + [100{,}000/(1.1)^5]$$

$$= -\$200{,}000 + \$18{,}181.82 + \$33{,}057.85 + \$45{,}078.89 + \$54{,}641.07 + \$62{,}092.13$$

$$= \$13{,}051.76$$

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